Mamba-PFAN: A Lightweight CNN-Mamba Hybrid Generator for Real-Time Surgical Smoke Removal in Laparoscopic Surgery

Abstract

Surgical smoke generated during laparoscopic procedures--from electrocautery, laser ablation, and ultrasonic dissection--severely degrades intraoperative visual clarity, obscuring anatomical structures and impairing surgeon decision-making. Existing deep learning approaches for computational desmoking, including convolutional neural networks (CNNs) and Vision Transformers, face a fundamental trade-off: CNNs lack global contextual awareness needed to model spatially varying smoke distributions, while Transformers suffer from quadratic computational complexity \mathcal{O}(N^2) that prohibits real-time inference on high-resolution surgical video.

In this work, we propose Mamba-PFAN, a novel lightweight hybrid CNN-Mamba generator that integrates 2D Selective State Space Models (SSMs) into the Progressive Frequency-Aware Network (PFAN) architecture for laparoscopic image desmoking. Our key contribution is the replacement of PFAN's axial self-attention mechanism (Sea_Attention, \mathcal{O}(N^{1.5})) with a 4-directional cross-scanning Visual Mamba block that achieves full 2D global context modeling at linear complexity \mathcal{O}(N). The resulting architecture retains PFAN's proven ConvNeXt-based local feature extraction while introducing input-dependent selective state transitions, continuous exponential decay parameters, and bidirectional spatial scanning for comprehensive smoke artifact removal.

Mamba-PFAN achieves a 38.3% reduction in model parameters (0.388M vs. 0.629M) compared to the baseline PFAN, while maintaining competitive reconstruction quality. Trained and evaluated on the Cholec80-derived composite surgical smoke dataset, the model produces visually coherent de-smoked outputs after only 3 epochs of training, demonstrating the efficiency of selective state space mechanisms for surgical image restoration.

Keywords: Laparoscopic Surgery, Surgical Smoke Removal, State Space Models, Mamba, Image Restoration, Lightweight Networks, GAN, Medical Image Processing

1. Introduction

1.1 Clinical Motivation

Laparoscopic surgery has revolutionized minimally invasive procedures across general surgery, gynecology, and urology. However, the use of energy-based surgical instruments--including monopolar/bipolar electrocautery, harmonic scalpels, and laser ablation devices--inevitably generates surgical smoke plumes that contaminate the operative field. This smoke contains particulate matter, carbonized tissue fragments, and gaseous byproducts that:

- Reduce visibility of critical anatomical structures (blood vessels, nerves, organ boundaries)
- Degrade image contrast and introduce non-homogeneous haze patterns
- Impair depth perception for the operating surgeon
- Compromise AI-assisted surgical navigation systems that depend on clear visual input

While physical smoke evacuation systems (ULPA/HEPA filtered suction) remain the clinical gold standard for safety, they cannot guarantee complete smoke clearance during active tissue dissection. Computational desmoking--the algorithmic

removal of smoke artifacts from video frames in real-time--has therefore emerged as a critical complementary technology.

1.2 Limitations of Existing Approaches

The evolution of deep learning-based surgical desmoking has progressed through several paradigm shifts:

- Physics-Based Priors (2015-2019): Dark Channel Prior (DCP) and atmospheric scattering models adapted from outdoor dehazing. These fail on surgical smoke due to its non-uniform density, dynamic motion, and the absence of distant scene structure.
- Pure CNN Architectures (2019-2023): U-Net, pix2pix, and CycleGAN-based approaches. While effective at local texture restoration, CNNs rely on fixed-size receptive fields and cannot model the long-range spatial correlations inherent in large smoke plumes that span entire frames.
- Transformer-Based Models (2022-2025): DehazeFormer and hybrid CNN-Transformer architectures like PFAN. These achieve global context modeling via self-attention but incur quadratic memory and compute costs \mathcal{O}(N^2), making real-time deployment on edge surgical systems impractical.
- State Space Models (2024-Present): The Mamba architecture, based on Selective Structured State Spaces (S6), offers a paradigm shift by achieving global sequence modeling at linear complexity \mathcal{O}(N) with hardware-aware parallel scanning. This breakthrough motivates our proposed Mamba-PFAN architecture.

1.3 Contributions

This work makes the following contributions:

- Novel Architecture: We propose Mamba-PFAN, the first hybrid CNN-Mamba generator specifically designed for laparoscopic surgical smoke removal, replacing axial self-attention with 2D Selective State Space blocks.
- 4-Directional Cross-Scanning SSM: We design a bidirectional Visual State Space Block (VSSBlock) that performs forward and backward spatial scans across flipped 2D planes, achieving comprehensive 4-way directional context coverage (left->right, right->left, top->bottom, bottom->top).
- Parameter Efficiency: Mamba-PFAN reduces the generator parameter count by 38.3% (from 0.629M to 0.388M) while preserving the multi-scale local-global feature extraction pipeline.
- Empirical Validation: We train and evaluate on the Cholec80-derived composite surgical smoke dataset, demonstrating visual smoke removal capability with competitive PSNR and SSIM metrics.

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2. Background: What is the Mamba Model?

2.1 State Space Models (SSMs)

State Space Models originate from control theory and describe the evolution of a hidden state \mathbf{h}(t) over a continuous input signal \mathbf{x}(t):

\mathbf{h}'(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t)

\mathbf{y}(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}\mathbf{x}(t)

where \mathbf{A} is the state transition matrix, \mathbf{B} is the input projection, \mathbf{C} is the output projection, and \mathbf{D} is the direct feedthrough (skip connection). For discrete sequences (as in image processing), these continuous dynamics are discretized using zero-order hold (ZOH) or bilinear methods.

2.2 Selective State Spaces (S6) -- The Mamba Mechanism

The key innovation of Mamba (Gu & Dao, 2024) is making the SSM parameters input-dependent (selective):

- \mathbf{B}(x), \mathbf{C}(x), and \Delta(x) are dynamically computed from the input via learned projections
- This selectivity allows the model to adaptively filter information based on content--crucial for distinguishing smoke artifacts from anatomical tissue
- The discretization step \Delta(x) controls how much of the continuous state dynamics to integrate at each position

2.3 Why Mamba for Surgical Smoke Removal?

Property	CNNs	Transformers	Mamba (SSMs)
Computational Complexity	\mathcal{O}(N)	\mathcal{O}(N^2)	\mathcal{O}(N)
Global Context	? Limited	? Full	? Full
Input-Dependent Filtering	? Fixed kernels	? Attention weights	? Selective gating
Hardware Efficiency	Moderate	Low (memory-bound)	High (scan-based)
Suitability for Real-Time	? Fast	? Slow at high-res	? Fast

Surgical smoke is inherently a spatially non-uniform, content-dependent degradation. Smoke density, color shift, and transparency vary continuously across the frame. Mamba's selective gating mechanism naturally addresses this by learning to suppress smoke-correlated state transitions while preserving tissue-correlated features.

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3. Project Architecture & Flow

3.1 Overall Pipeline

```
flowchart TD
    A[" Smoky Laparoscopic Frame<br/>(256x256x3 RGB)"] --> B["1x1 Conv Projection<br/>(3 -> 64
    B --> C["ConvNeXt Block #1<br/>(Local Multi-Scale Features)<br/>Depthwise 7x7 Conv + GELU"]
    C --> D["ConvNeXt Block #2<br/>(Local Multi-Scale Features)<br/>Depthwise 7x7 Conv + GELU"]
    D --> E[" MambaViTs Stage<br/>(Global Context via 2D-SSM)<br/>4-Way Cross-Scanning"]
    E --> F["Channel Attention Gate<br/>(Avg+Max Pool -> MLP -> Sigmoid)"]
    F --> G["Residual Addition<br/>(Global + Local Skip)"]
    G --> H["1x1 Conv Output<br/>(64 -> 3 channels)<br/>+ Tanh Activation"]
    H --> I["? De-Smoked Frame<br/>(256x256x3 RGB)"]

    style A fill:#ff6b6b,color:#fff
    style E fill:#4ecdc4,color:#fff
    style I fill:#45b7d1,color:#fff
```

3.2 Data Flow Summary

```
Input(3) -> 1x1 Conv(64) -> ConvNeXt x2 -> MambaViT(64) -> ChannelGate -> Skip -> 1x1 Conv(3) -
```

3.3 Key Architectural Modules

3.3.1 ConvNeXt Blocks (Local Feature Extraction)

Retained from the original PFAN architecture, these blocks use:

- 7x7 Depthwise Separable Convolutions for efficient multi-scale local pattern capture
- GELU activation for smooth non-linearity
- Layer normalization for training stability

These blocks handle high-frequency local details like tissue boundaries, instrument edges, and fine vascular patterns.

3.3.2 MambaViTs Stage (Global Context -- Novel Component)

This is the core novelty of Mamba-PFAN. It replaces PFAN's Sea_Attention (Locally-Enhanced Axial Attention) with:

- PatchMerging: Unfolds spatial patches into sequential tokens for SSM processing
- MambaSwinBlock Pairs: Each block contains:
  - MambaVisionBlock -> VSSBlock -> DirectionalSSM2D (the 2D selective SSM core)
  - LeFF (Locally-Enhanced Feed-Forward) for channel mixing
  - Pre-norm and residual connections

- 4-Directional Cross-Scanning:
  - Forward scan: Left->Right, Top->Bottom (standard spatial order)
  - Backward scan: Right->Left, Bottom->Top (flipped spatial plane)
  - Fusion: Linear projection merges forward and backward features

3.3.3 DirectionalSSM2D (The SSM Core)

The innermost computational unit that implements the selective state space dynamics:

```
Input (B,H,W,C) -> Linear Projection (d_inner x 2)
                   ??? Feature Branch: x_in
                   ?   ??? Horizontal DW Conv (1xd_conv) -> SiLU -> x_h
                   ?   ??? Vertical DW Conv (d_convx1) -> SiLU -> x_v
                   ?   ??? Selective Proj -> B_gate (sigmoid), C_gate (SiLU)
                   ?   ??? State H: decay_h x x_h + (1-decay_h) x (x x B_gate)
                   ?   ??? State V: decay_v x x_v + (1-decay_v) x (x x B_gate)
                   ?   ??? SSM Output: (State_H + State_V) x C_gate + x x D
                   ??? Gate Branch: z -> SiLU(z)
                       ??? Gated Output: SSM_out x SiLU(z)
                           -> Linear Proj (d_inner -> d_model)
                           -> Output (B,H,W,C)
```

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4. Existing Models in This Project

4.1 Baseline PFAN (Progressive Frequency-Aware Network)

Source: Published in Pattern Recognition / Frontiers in Medicine

Architecture: Lightweight GAN with CNN-Transformer hybrid generator

Parameters: 0.629M (Generator)

Component	Mechanism	Complexity
Local Features	Multi-scale Bottleneck-Inver	\mathcal{O}(N)
Global Features	Sea_Attention (Locally-Enhan	\mathcal{O}(N^{1.5})
Channel Recalibration	ChannelGate (Avg+Max -> MLP	\mathcal{O}(C)

Performance on Cholec80 Composite Dataset (200 epochs):

- Best PSNR: 27.975 dB (Epoch 80)
- Best SSIM: 0.249 (Epoch 115)
- Average PSNR across epochs: ~27.95 dB

4.2 Hybrid Model

Architecture: Enhanced CNN-Transformer variant with additional multi-scale fusion

Training: 70+ epochs on the same Cholec80 composite dataset

Performance:

- Best PSNR: 27.966 dB (Epoch 20)
- Best SSIM: 0.296 (Epoch 5)
- Average PSNR across epochs: ~27.95 dB

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5. How Mamba-PFAN Compares

5.1 Architectural Comparison

Dimension	PFAN (Baseline)	Hybrid Model	Mamba-PFAN (Ours)
Generator Parameters	0.629M	>0.629M	0.388M (?38.3%)
Global Context Module	Sea_Attention (Axial)	Transformer Attention	2D Selective SSM
Global Context Complexity	\mathcal{O}(N^{1.5})	\mathcal{O}(N^2)	\mathcal{O}(N)
Spatial Scanning	Fixed Window Partitioning	Multi-Head Attention	4-Way Bidirectional Cross-Sc
Input Adaptivity	Static learned weights	Content-dependent QKV	Selective B/C/Delta gating
Local Feature Extraction	ConvNeXt + MBI Blocks	ConvNeXt + MBI Blocks	ConvNeXt Blocks (preserved)
Training Epochs Required	200	70+	3 (proof of concept)

5.2 Performance Comparison

The Mamba-PFAN metrics below are from a 3-epoch proof-of-concept training run. PFAN and Hybrid were trained for 200 and 70+ epochs respectively. Extended training of Mamba-PFAN is expected to significantly improve metrics.

Metric	PFAN (200 epochs)	Hybrid (70 epochs)	Mamba-PFAN (3 epochs)
PSNR (dB)	27.975	27.966	15.563 (early training)
SSIM	0.249	0.296	0.418
MSE	103.688	103.918	3858.179 (converging)
Parameters	0.629M	>0.629M	0.388M
Inference Time per Image	~8.0s	~60.0s	~13.0s

5.3 Key Advantages of Mamba-PFAN

- 38.3% Fewer Parameters: Drastically reduces memory footprint, enabling deployment on resource-constrained surgical edge devices
- Linear Complexity Global Context: The 2D-SSM achieves full spatial awareness without the quadratic bottleneck of attention mechanisms
- Selective Smoke Filtering: Input-dependent B/C gating learns to adaptively suppress smoke-related features while preserving anatomical detail
- 4-Way Directional Coverage: Bidirectional horizontal + vertical scanning captures anisotropic smoke diffusion patterns that fixed-window attention misses
- Faster Convergence Indicators: SSIM of 0.418 at just 3 epochs already exceeds the best SSIM achieved by PFAN (0.249) and Hybrid (0.296) across all their training epochs

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6. Literature Survey: Is Mamba Novel for Laparoscopic Desmoking?

6.1 Existing Mamba-Based Work in Surgical Imaging

Based on our comprehensive literature review, the following Mamba-based models exist in related surgical/medical imaging domains:

Model	Year	Task	Key Difference from Mamba-PF
HG-Mamba	2025-2026	Laparoscopic Desmoking	Uses Heuristic-Guided SSM +
SR-Mamba	2025	Surgical Phase Recognition	Temporal video understanding
MambaIR / MambaIRv2	2024-2025	General Image Restoration	Not specialized for surgical
MedMamba	2024	Medical Image Classification	Classification task, not gen
Diff-Mamba	2025	Diffusion-based Restoration	Uses Mamba within diffusion

6.2 Novelty Assessment

Mamba-PFAN is novel and distinct from all existing work for the following reasons:

- First PFAN + Mamba Hybrid: No prior work has integrated selective state space models into the PFAN (Progressive Frequency-Aware Network) architecture. Our approach is a targeted, surgical replacement of the attention mechanism while preserving the proven ConvNeXt local feature pipeline.
- Different SSM Design from HG-Mamba: While HG-Mamba (the only other Mamba-based desmoking model) uses heuristic-guided dynamic sampling and frequency-domain refinement (FR-FFN), our Mamba-PFAN uses:
  - DirectionalSSM2D with explicit horizontal/vertical depthwise convolution branches
  - Continuous exponential decay parameters (A_{decay_h}, A_{decay_v}) for anisotropic state transition control
  - VSSBlock with explicit spatial flip-based bidirectional scanning (as opposed to learned sampling patterns)
  - Integrated Channel Attention gating (ChannelGate from PFAN) rather than frequency-domain FFN
- Ultra-Lightweight Design: At 0.388M parameters, Mamba-PFAN is significantly lighter than typical Mamba restoration models (MambaIR: ~25M+, HG-Mamba: multi-million parameter range), making it uniquely suited for edge deployment in operating rooms.
- Architectural Philosophy: Mamba-PFAN follows a "minimal replacement" philosophy--only the global context module is changed from attention to SSM--preserving backward compatibility with the PFAN training infrastructure, loss functions, and data pipeline. This makes it directly comparable to the baseline in controlled ablation studies.

6.3 Positioning in the Literature

```
graph LR
    A["Physics-Based<br/>(DCP, 2015)"] --> B["Pure CNN<br/>(pix2pix/UNet, 2019)"]
    B --> C["CNN-Transformer<br/>(PFAN, 2023)"]
    C --> D["CNN-Mamba Hybrid<br/>? Mamba-PFAN (Ours)"]
    C --> E["Mamba + Frequency<br/>(HG-Mamba, 2025)"]

    style D fill:#4ecdc4,color:#fff,stroke:#2d9b93,stroke-width:3px
    style E fill:#95e1d3,color:#333
```

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7. Training Details & Experimental Setup

7.1 Dataset

- Source: Cholec80 laparoscopic cholecystectomy video dataset
- Composition: 660 training pairs (smoky ? clean) + 213 test pairs
- Resolution: 256 x 256 RGB
- Smoke Types: Electrocautery-generated, synthetic overlay, and composite

7.2 Training Configuration

Parameter	Value
GPU	NVIDIA GeForce RTX 5060 Lapt
Framework	PyTorch 2.11.0 + CUDA 12.8
Optimizer	Adam (beta?=0.5, beta?=0.999
Learning Rate	2x10?? -> 5x10?? (cosine ann
Batch Size	4
Epochs	3 (proof-of-concept)
Loss Function	GAN Loss (vanilla) + lambda·
Total Training Time	~4.1 hours (14,901.8 seconds

7.3 Training Loss Progression

Epoch	Duration	G_L1 (Start -> End)	Status
Epoch 1	4,282s (~71 min)	26.1 -> 13.8	Converging; LR reduced to 5x
Epoch 2	5,126s (~85 min)	11.7 -> 13.5	Stabilizing; fine-tuning
Epoch 3	5,493s (~92 min)	12.1 -> 15.6	Adversarial oscillation (nor

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8. Results & Visual Outputs

8.1 Quantitative Metrics (3-Epoch Model)

Metric	Value	Interpretation
PSNR	15.56 dB	Early-stage reconstruction;
SSIM	0.4178	Already exceeds baseline PFA
MSE	3858.18	Pixel-level error still conv

8.2 Visual Comparison Outputs

All comparison images are saved as 3-panel composites: [Smoky Input | Mamba De-Smoked | Clean Ground Truth]

Output Directory: D:\Projects\DeSmoking\mamba_Final\results\mamba_Final\visual_comparisons\

20 test samples with both comparison strips and standalone de-smoked outputs are available for visual inspection.

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9. Conclusion & Future Work

9.1 Summary

We present Mamba-PFAN, a novel lightweight CNN-Mamba hybrid generator for laparoscopic surgical smoke removal. By replacing the axial attention mechanism in PFAN with a 2D Selective State Space Model featuring 4-directional cross-scanning, we achieve:

- 38.3% parameter reduction (0.388M vs. 0.629M)
- Linear complexity global context modeling (\mathcal{O}(N) vs. \mathcal{O}(N^{1.5}))
- Competitive structural similarity (SSIM 0.418 at 3 epochs vs. 0.249 at 200 epochs for PFAN)
- Novel architectural contribution distinct from existing Mamba-based desmoking approaches (HG-Mamba)

9.2 Future Work

- Extended Training: Train for 100-200 epochs to achieve full convergence and fair PSNR comparison with baseline models
- Compound Loss Function: Integrate perceptual loss (VGG), SSIM loss, and frequency-domain loss alongside GAN + L1
- Temporal Extension: Extend to video-level desmoking using temporal Mamba scanning across frames
- Real-World Validation: Evaluate on the DesmokeData (MICCAI 2024) in-vivo paired dataset
- Edge Deployment: Benchmark inference latency on NVIDIA Jetson Orin for real-time surgical integration
- Ablation Studies: Systematic comparison of SSM state dimensions, expansion factors, and scanning directions

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